SDGs, Ethics & Philosophy

Lecture 3 Retake

AI for Good - Week 3

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THREE PRINCIPLES FOR SAFER AI

Stuart Russell

Only goal: make human values come true

Uncertain about what those values are

Learns our values by watching human behaviour

Uncertainty means it lets you switch it off



THE SDGS AND MY SDG

17 goals, all UN countries

2015 to 2030, 169 targets

People, planet, prosperity, peace, partnership

My SDG: 3
Good Health and Well-being

My hackathons on medication and lab results; health is the basis for school, work and society



AI FOR SDG 3

EYE SCREENING IN THAILAND

Google AI checks eye photos
for diabetic retinopathy

Promise: over 90% accurate
in the lab, no waiting for a specialist

Problem: about 1 in 5 photos
rejected (bad light), slow internet

Lab success is not real-life success



EU ETHICS GUIDELINES FOR TRUSTWORTHY AI

Human agency and oversight

Technical robustness and safety

Privacy and data governance

Transparency

Diversity, non-discrimination and fairness

Societal and environmental well-being

Accountability



TWO REQUIREMENTS IN PRACTICE

HUMAN OVERSIGHT

A doctor or nurse reviews
the AI output and can overrule it

ROBUSTNESS

Test in real clinics first
Have a backup plan when
a photo is rejected



THREE PHILOSOPHICAL LENSES



Utilitarianism

Greatest good for the greatest number

Deontology

Rules, rights and duties, whatever the outcome

Virtue ethics

What would a person of good character do?

EVALUATING BE MY AI

Be My Eyes app, since 2023:
GPT-4 describes photos for
blind and low vision people

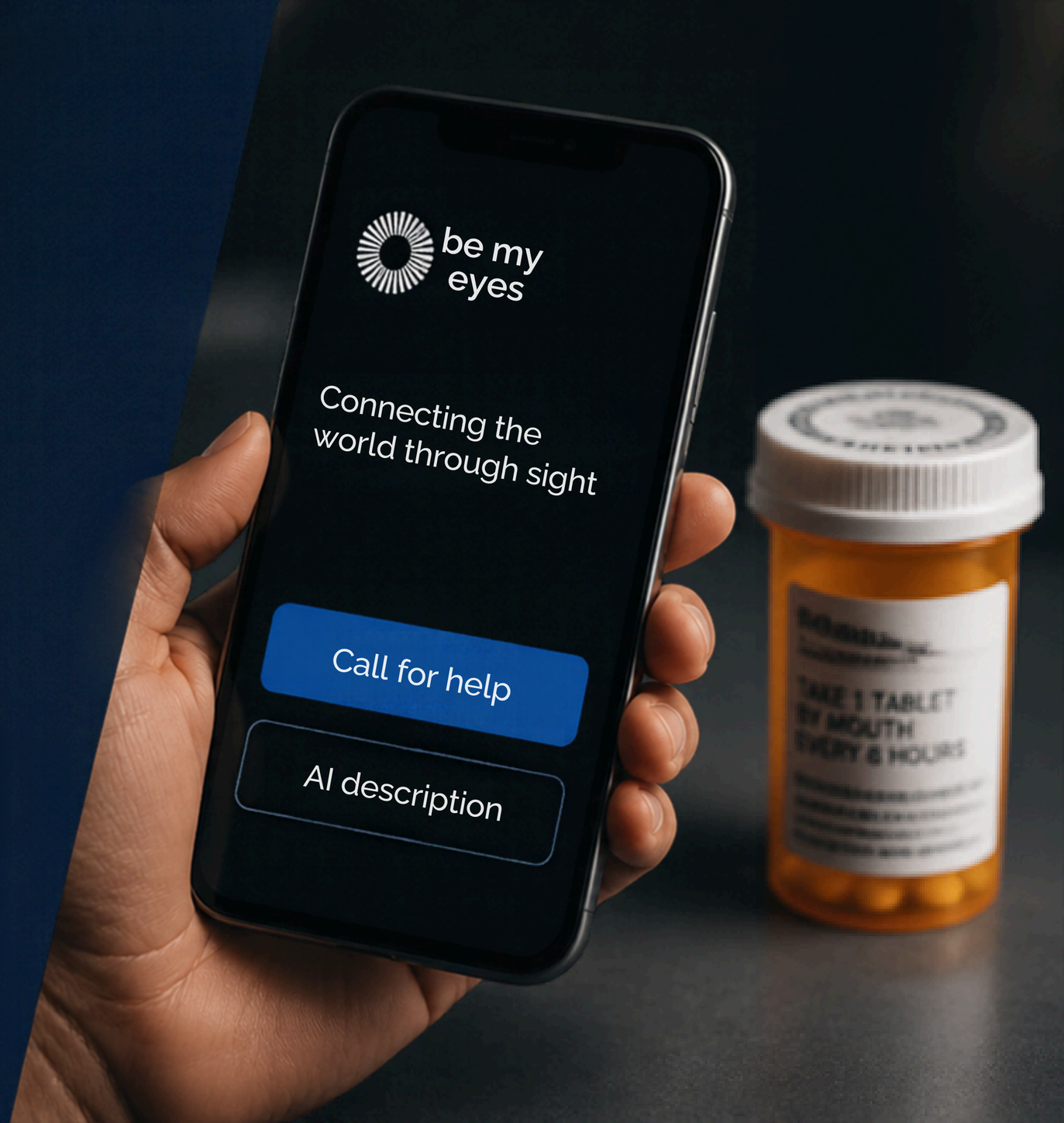
Goal: more independence

PRIVACY

photos of homes,
documents and
other people

ACCOUNTABILITY

who is responsible
for a wrong
medicine label?



THREE LENSES AND MY VERDICT

UTILITARIAN

helps many people every day

DEONTOLOGY

right to know where photos go; people in photos did not consent

VIRTUE

be honest that AI makes mistakes



VERDICT: ACCEPTABLE UNDER CONDITIONS

clear warnings

human volunteer one click away

double-check medicine

protect photos

THANK YOU

Lecture 3 slides

Stuart Russell, TED 2017

European Commission, Ethics Guidelines for Trustworthy AI (2019)

Google Health field study in Thailand (2020)

Be My Eyes (2023)

